# Applied Machine Learning for Business Analytics

Lecture 8: Tool

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# Agenda

1. Tools
2. MCP

# 1. Tools

# Tools

- In nature, LLM is only able to do next token prediction
  - A brilliant brain with no hands
- By enabling external tools, agents can have more capabilities.
- Tools can be classified into two types:
  - Reading
    - SQL executor, API call & web crawl-> have more accurate and up-to-date info
    - Calculator, unit converters, code interpreters, unit converters -> reduce hallucination
  - Writing
    - Modify data sources, email API sending responses, modify code etc -> enable automation in workflow

**Tool calling** is how we give the LLM hands — letting it trigger real actions in the real world, while the LLM stays in charge of *deciding what to do*.

# Tool Calling

- Tools: functions made available to LLMs
- How to make it available to LLM: Tool Schema
  - An appropriate name
  - Relevant Parameters
  - Description of the tool's purpose
- Tool Calling
  - LLMs **do not execute the function**, it only generates a structured schema for the tool specifying
    - Tool name with the necessary parameters -> basic info to execute the tools

# How the LLM knows when & how to use a tool

**A tool definition looks like this**

```
name: "get_weather"  ← identifier
 description: "Get current weather for a city"
← WHEN to call it
 parameters: {
  location: string
  ← WHAT to pass
}
```

**Description = instruction to the model**

The LLM reads the description to decide whether to call a tool. Write clear, specific descriptions — they are instructions for the model, not docs for humans.

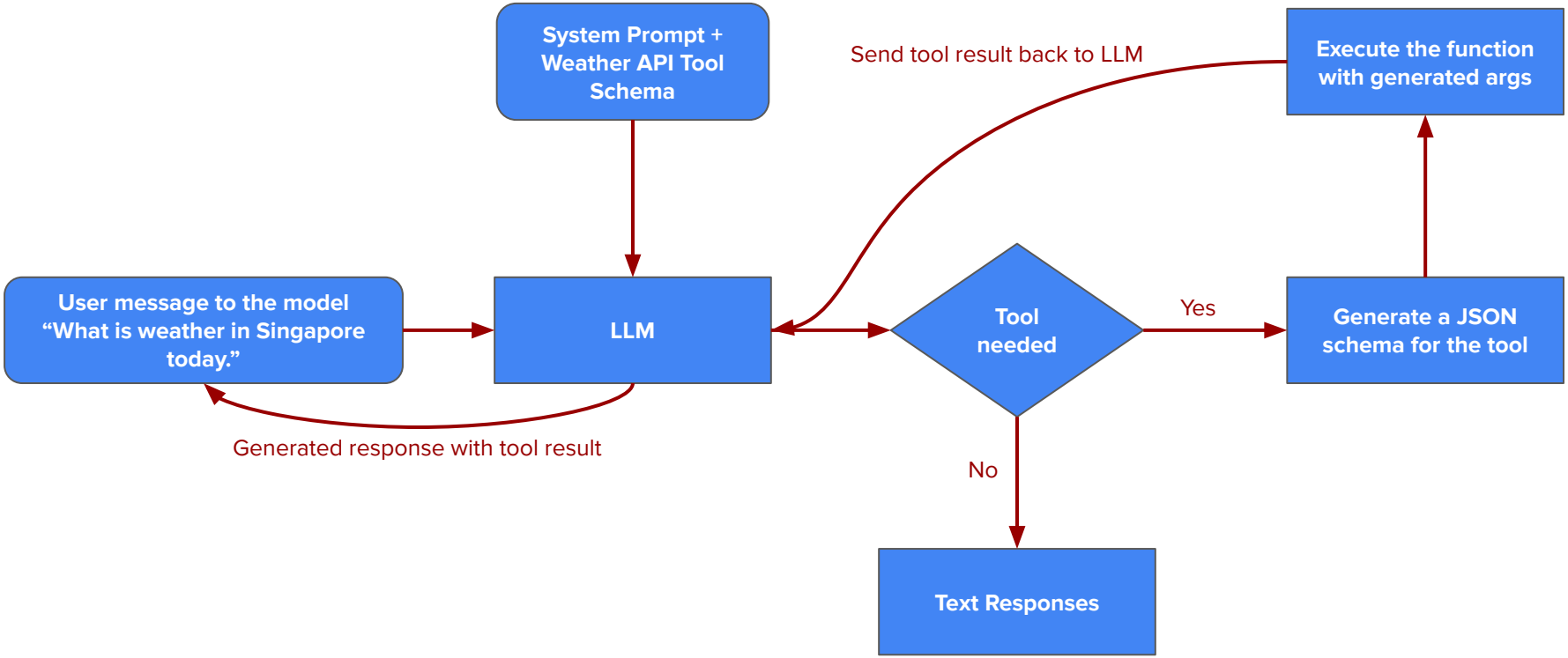
**Parameters are filled from the conversation**

location = "Singapore" is inferred automatically from the user's message and the parameter type.

**No match → plain text, no forced call**

If no tool description fits the question, the LLM answers normally in plain text.

# Tool Calling



# 5-step loop - tool calling

Step 1

💬

**User asks**

Natural language question or instruction

→

Step 2

🤔

**LLM picks a tool & responds**

Decides which tool fits and what parameters to pass

→

Step 3

⚙️

**App runs it**

Your application executes the tool — API call, DB query, etc.

→

Step 4

📬

**Result returns**

Tool output is fed back to the LLM as context

→

Step 5

✅

**LLM answers**

Uses the real data to compose a grounded, accurate response

**The LLM never *runs* the tool itself — it only *decides* to use one. Your application does the actual work and feeds the result back.**

# Example: what is the weather in Singapore

**Step 1**

💬

User asks

What is the weather in Singapore?

→

**Step 2**

🤔

LLM picks a tool and responds

get_weather
location: Singapore

→

**Step 3**

⚙️

App runs it

Hits weather API with location-Singapore

→

**Step 4**

📬

Result returns

partly cloudy

→

**Step 5**

✅

LLM answers

It is partly cloudy in Singapore right now

```
// What the LLM actually returns in Step 2:
{ "tool": "get_weather", "arguments": { "location": "Singapore" } }
```

# How LLM have the Tool Calling Capability

## Baked into the model via post-training

Modern LLMs (GPT-4, Claude, Gemini...) are specifically trained to:

✓ Read tool schemas — name, description, parameters

✓ Decide when a question needs a tool vs. a plain text answer

✓ Respond with structured tool calls (name + arguments) instead of string (usually json format)

*Not a plugin or add-on — a core part of how the model works.*

## Activated per request — not globally

There is no "tool mode" switch. You activate it for a specific request by including tool definitions in that API call.

✓ Include tool definitions → LLM can return tool calls

✓ Omit them → LLM answers in plain text only

✓ Each request independently controls which tools are available

*No extra config — just include the list in your API call.*

Enabling tool calling = **a model trained for it** + **tool definitions included in the prompt**.

# Training Samples for Tool Calling

Standard CHAT

```
Messages = [
{ "role": "user",
  "content": "help with order #123”},
{ "role": "assistant",
  "content": "I would be happy to help”},
]
```

Tool-calling CHAT

```
Messages = [
{ "role": "user",
  "content": "Status of order #123”},
{ "role": "assistant",
  "tool_call”: {
   “name”: “get_status”,
   “arguments”: ‘{“txn”:”T1001”}’}},
{ "role": "tool",
  "content": ‘{“status”:”paid”}’},
{ "role": "assistant",
  "content": "Transaction #123: Paid”},
]
```

Training samples must demonstrate **the complete cycle**. The model learns to: decide which tool to call, format the call correctly, then interpret the result.

Tutorial: https://huggingface.co/learn/agents-course/en/bonus-unit1/introduction

# Prompt Template

Identity

You are a helpful assistant. You should think step by step.
Use tools only when they are necessary for the task. If a query can be answered directly, respond with a simple message instead of using tools. You have the following tools available to you.
Tools: [...]
A few shots: [(query;tool_calls), ....]

Instructions

# Langchain’s Tool Calling: Big Picture

- In LangChain, we do not need to write the system prompt for tool calling manually. It is constructed automatically from two things we provide

**Tool Definitions**

Python functions decorated with @tool - their names, docstrings, and type hints.

**bind_tools()**

Seralizes those definitions into JSON Schema and injects them into every API call

# Define Tools via Decorator

**The @tool decorator reads three parts of the function and converts them into a JSON Schema the model can understand**

```python
@tool
def get_current_weather(location: str, unit: str = "celsius") -> str:
    """Get current weather in a location."""
    ...

@tool
def calculate(expression: str) -> str:
    """Evaluate a math expression."""
    ...
```

Auto-generated JSON Schema

```
{
  "name": "get_current_weather",
  "description": "Get current
    weather in a location.",
  "parameters": {
    "location": {
      "type": "string"
    },
    "unit": {
      "type": "string",
      "default": "celsius"
    }
  }
}
```

LLMs uses func. Name to call the right tool

Description of what the tool does

Validate LLM' argument output

LLM can omit optional params.

**Docstrings are the documentation for LLM. The LLM never reads the source code - only the schema. Write docstring as if briefing the LLM, not just colleagues**

# Create Hidden System Prompt: bind_tools()

```
model_with_tools = model.bind_tools(TOOLS)
```

Auto-Inject

You have access to these tools:

Tool: get_current_weather
  Description: Get current weather in a location.
  Parameters: { location: string, unit: string (default celsius) }

Tool: calculate
  Description: Evaluate a math expression.
  Parameters: { expression: string }

When calling a tool, respond with JSON only:
{"tool_name": "...", "arguments": {...}}

Usually invisible

**Bind_tools() serializes all tool schemas and injects them into the LLM API call as special tools parameters (OpenAI) or an equivalent system-level prompt for other LLMs**

# OpenAI and Antropic

OpenAI

```python
response = client.responses.create(
    model="gpt-5",
    instructions="Respond only with a horoscope generated by a tool.",
    tools=tools,
    input=input_list,
)
```

**Auto-Inject tool info into prompt for LLM call**

Antropic

```python
import anthropic

client = anthropic.Anthropic()

response = client.messages.create(
    model="claude-opus-4-6",
    max_tokens=1024,
    tools=[
        {
            "name": "get_weather",
            "description": "Get the current weather in a given location",
            "input_schema": {
                "type": "object",
                "properties": {
                    "location": {
                        "type": "string",
                        "description": "The city and state, e.g. San Francisco, CA",
                    }
                },
                "required": ["location"],
            },
        }
    ],
    messages=[{"role": "user", "content": "What's the weather like in San Francisco?"}],
)
print(response)
```

# The Full Prompt

```
messages = [HumanMessage(content=question)]
```

Auto-injected Tool Schema Block

```
You have access to these tools:

Tool: get_current_weather
  Description: Get current weather in a location.
  Parameters: { location: string, unit: string (default celsius) }

Tool: calculate
  Description: Evaluate a math expression.
  Parameters: { expression: string }

When calling a tool, respond with JSON only:
{"tool_name": "...", "arguments": {...}}
```

HumanMessage

```
What is the weather like in Singapore?
```

# Optional Custom System Message

- Customizing system prompt
  - Role definition - who is the assistant?
  - Goal and constraints - when should it use tools?
  - Few-shot examples - show example tool calls?

```python
from langchain_core.messages import SystemMessage

messages = [
    SystemMessage(content="""You are a helpful assistant.
Always use the available tools when the question requires real data.
If the user asks about weather, use get_current_weather.
If the user asks for a calculation, use calculate."""),
    HumanMessage(content=question)
]
```

You have access to these tools:

Tool: get_current_weather
  Description: Get current weather in a location.
  Parameters: { location: string, unit: string (default celsius) }

Tool: calculate
  Description: Evaluate a math expression.
  Parameters: { expression: string }

When calling a tool, respond with JSON only:
{"tool_name": "...", "arguments": {...}}

You are a helpful assistant.
Always use the available tools when the question requires real data.
If users ask about weather, use get_current weather,
If users ask for a calculation, use calculate

What is the weather like in Singapore?

# Tool Callings Loop

Step 2

## Send query + Tools

HumanMessage + Tool Schemas sent to model

```
messages = [HumanMessage(content=question)]

for _ in range(max_rounds):
    response = model_with_tools.invoke(messages)
```

Step 3

## Model decides

Does response.tool_calls have entries?

- no->final answer
- yes -> continue

```
response = model_with_tools.invoke(messages)
print("Model:", response.content)

# STEP 3: If no tool calls, we have a final answer
if not response.tool_calls:
    print("Answer:", response.content or "(no content)")
    return

messages.append(response)
```

Step 4

## Execute tool

The Python code calls run_tool(func_name, arg) - LLM is NOT involved here

```
# STEP 4: Execute each tool call in application code
for tool_call in response.tool_calls:
    name = tool_call["name"]
    args = tool_call.get("args", {})
    print(f"Tool requested -> {name}({args})")
    tool_result = run_tool(name, args)
```

Step 5

## Return Result

Toolmessage appended to conversion. LLM reads it and gives a final answer.

```
# STEP 5: Send tool result back to the LLM via ToolMessage
messages.append(
    ToolMessage(content=tool_result, tool_call_id=tool_call["id"])
)
```

# Message History as the loop runs

```
# Initial
messages = [HumanMessage("What's the weather in London?")]

# After model responds with tool_calls
messages += [AIMessage(tool_calls=[{"name": "get_current_weather", "args": {"location": "London"}}])]

# After your code executes the tool
messages += [ToolMessage("London: 15.0C, sunny", tool_call_id="...")]

# Model now has context to give final answer
```

# Why the Docstring Matters

```python
@tool
def get_current_weather(location: str, unit: str = "celsius") -> str:
    """Get current weather in a location."""
    ...

@tool
def calculate(expression: str) -> str:
    """Evaluate a math expression."""
    ...
```

This is ALL the model knows about this tool

| Bad | Good |
|---|---|
| “”” Does weather stuff.”””  | “”” Get current weather in a location. Returns temperature and conditions”””  |
| “”” Math.”””  | “”” Evaluate a math expression like ‘4*7/3’. Returns the numeric result.”””  |

# Tool Callings: Yes

```
result = model_with_tools.invoke("What is the weather like in Singapore")
print(result)
```

Generate a JSON schema for the tool

**content**=''
**additional_kwargs**={}
**response_metadata**={'*model*': 'qwen2.5:3b', '*created_at*': '2026-03-07T12:08:30.513783Z', '*done*': True, '*done_reason*': 'stop', '*total_duration*': 654724833, '*load_duration*': 30845750, '*prompt_eval_count*': 215, '*prompt_eval_duration*': 184000000, '*eval_count*': 21, '*eval_duration*': 437000000, '*logprobs*': None, '*model_name*': 'qwen2.5:3b'}
**id**='run--019cc832-f9a1-7b31-beca-a5571dcea975-0'
**tool_calls**=[{'*name*': 'get_current_weather', '*args*': {'location': 'Singapore'}, '*id*': 'd759987e-f8e8-497f-8233-26aabc68e149', '*type*': 'tool_call'}]
**usage_metadata**={'*input_tokens*': 215, '*output_tokens*': 21, '*total_tokens*': 236}

```python
# STEP 4: Execute each tool call in application code
for tool_call in response.tool_calls:
    name = tool_call["name"]
    args = tool_call.get("args", {})
    print(f"Tool requested -> {name}({args})")
    tool_result = run_tool(name, args)

    # STEP 5: Send tool result back to the LLM via ToolMessage
    messages.append(
        ToolMessage(content=tool_result, tool_call_id=tool_call["id"])
    )
```

# Tool Callings: No

```
result = model_with_tools.invoke("Hello, how are you?")
print(result)
```

```
# STEP 3: If no tool calls, we have a final answer
if not response.tool_calls:
    print("Answer:", response.content or "(no content)")
    return
```

content="I'm here to help with any questions or tasks you have. How can I assist you today?"
additional_kwargs={}
response_metadata={'model': 'qwen2.5:3b', 'created_at': '2026-03-07T12:08:31.634181Z', 'done': True, 'done_reason': 'stop', 'total_duration': 537898250, 'load_duration': 12316459, 'prompt_eval_count': 213, 'prompt_eval_duration': 93000000, 'eval_count': 21, 'eval_duration': 430000000, 'logprobs': None, 'model_name': 'qwen2.5:3b'}
id='run--019cc832-fe76-70c0-90c8-6fd348e8a280-0'
usage_metadata={'input_tokens': 213, 'output_tokens': 21, 'total_tokens': 234}

# Common Misconceptions

| Myth | Fact |
| --- | --- |
| I need to write a system prompt listing my tools | bind_tools() injects the tool description automatically |
| Docstrings are just for developers | LLM never read source code. Only the schema +doctstrings is sent. |
| LLM executes the tool | Your code executes tools |
| One round-trip is enough | User query->LLM tool request->tool results->LLM final answer |

# Best Practices for Tool Design

- Detailed Description
  - The description is the only way the model knows when and how to use a tool:
    - Be Verbose: write 3-4 sentences per description. Do not just say what it does; say when it should be used and when it should not
    - Explain Parameters: describe every param's meaning, expected format, and how it affects the outcome
    - Clarify Boundaries: if you have two similar tools (`search_email` and `search_files`), explicitly explain the difference in their description)
- Consolidate for "Selection Simplicity"
  - Avoid Tool Bloat: Having too many narrow tools (`get_user_name` and `get_user_email`and `get_user_id`) confuses the model
  - Group related Actions: create a single, more capable tool with an `action` parameter (a `manage_user` tool with an enum for `get_ifno`, `update` or `delete`). This reduces "selection ambiguity)

# Best Practices for Tool Design

- Use Meaningful Namespacing
  - Prefixing: if you application connects to multiple services, use prefixes (`slack_send_message` and `github_create_issue`). This makes the tool's origin and purpose instantly recognizable
- Avoid Overloading the Model with Too Many Tools
  - Start with 3-5 well-scoped tools. Only add more when we are confident the model is handling current tools well
- Favour Models With Native Tool Calling Support
  - Not all language models are equally good at tool calling. OpenAI models with function calling or Antropic's Claude with tool use capabilities

More details: https://platform.claude.com/docs/en/agents-and-tools/tool-use/implement-tool-use

# 2. MCP

# What if tools weren’t hard-coded?

### 📱 Hard-coded tools (direct)

Your LLM apps were using  a fixed set of tools written by your team. Adding a new tool or modifying a new tool means changing the code and redeploying. Every app that needs the same copy.

### 🧩 MCP: tools as plugins

Your LLM apps discovers tools from external servers at runtime — like installing apps on your phone. New tools = connect to a new server. No code change required.

**MCP (Model Context Protocol)** is an open standard that lets your LLM apps discover and run tools from external servers — without changing your application code each time a new tool is needed.

Think of it like: direct tool calling is a custom remote control you built for one specific TV. MCP is a **universal remote standard** — any device that follows the standard works with any compatible remote.

# MCP: same loop, two things change

**Step 1**

🔍

**Discover tools from MCP servers**

LLM apps ask MCP servers: what tools do you have?

→

**Step 2**

🤔

**LLM picks a tool & responds**

Same as before

→

**Step 3**

⚙️

**Route call to MCP server**

LLMs forwards to the server that owns that tool; server runs it

→

**Step 4**

📬

**Result returns**

Tool output is fed back to the LLM as context

→

**Step 5**

✅

**LLM answers**

Uses the real data to compose a grounded, accurate response

**MCP only change: tool discovery and tool execution. This protocol standardizes the interface of tools towards LLM: how tools are discovered, called, and how results are returned**

# Director tool calling vs MCP

| Dimension | Direct Tool Calling | With MCP |
|---|---|---|
| Where tools live | Hard-coded in your application | Hosted by external MCP servers (local or remote) |
| Who maintains them | You, the app developer | Server operator — your team, another team, or third party |
| Discovery | Static — known at build time | Dynamic — discovered at runtime |
| Reusability | Coupled to your app only | Any MCP-compatible client can connect and reuse |
| Complexity | Simpler — one process, no extra infra | More moving parts — servers must be running and reachable |
| Best for | Prototypes, small apps, tightly-coupled logic | Shared capabilities, multi-team, third-party integrations |

# MCP in action: a fin analyst

🧠 **LLM**

Picks tools · generate answers

↕ 2. sends tool schemas · 5. returns final answer

📊 **Portfolio MCP Server**

get_holdings
get_pnl_summary
get_trade_history

1. list tools
↔
4. call tool

🔧 **Your App**

discovers tools
routes calls
passes results

1. list tools
↔
4. call tool

📈 **Market Data MCP Server**

get_quote
get_historical_prices
get_fx_rate

Final answer ↓ · ↑ user question

👤 **User**

"What's the current value of my AAPL position?"

# MCP in action: a fin analyst

1. At startup: App discovers 6 tools from 2 MCP servers (3 portfolio tools + 3 market data tools)

2. App sends: user message "What's my AAPL position?" + all 6 tool schemas → LLM

3. LLM returns two tool calls: get_holdings(AAPL) and get_quote(AAPL)

4. App routes to the right server:
   📊 Portfolio → 150 shares @ avg $178.50
   📈 Market Data → $242.10, MTD +5.3%

✔ LLM generates: "You hold 150 shares of AAPL at an average cost of $178.50. Current price is $242.10 — position value ~$36,315, unrealized gain ~$9,540. Month-to-date AAPL is up 5.3%."

Your App wrote **zero lines** of portfolio or market data logic. It just discovered tools, routed calls, and passed results — the servers owned all the domain knowledge.

# When to use direct vs MCP

## Use Direct Tool Calling when...

- ✓ You're prototyping or building fast
- ✓ You have ≤ 5 tools and they're straightforward
- ✓ Tools are specific to this one app only
- ✓ You need lowest possible latency (no network hop)
- ✓ Tool logic is tightly coupled to your app's state

## Use MCP when...

- ✓ Multiple apps or teams need the same tools
- ✓ Different teams own different capabilities
- ✓ You want to add tools without redeploying your app
- ✓ You're integrating with third-party providers
- ✓ You want a shared standard, not ad-hoc wiring

**Practical rule:** Start with direct. Graduate to MCP when you find yourself copying the same tool logic across multiple apps — or when team boundaries demand it.

# Summary

- The 5-step loop is universal
  - ask->pick->run->return->answer
- LLM decides, app acts
  - The model never directly calls APIs or queries databases. It returns structured instructions that your code executes - then feeds the result back
- MCP is the standardisation layer
  - Tools live outside your app - discoverable, reusable, maintainable by difference teams
- Start simple; graduate when you need scale
  - Direct tool calling ships faster. MCP pays off when same capability needs to be shared across functions

Next Class: Agent: Thinking Patterns